Remarks COMPARISON WITH REAL ANALYSIS

(*i*) As a consequence of (4), we can say that the complex number system is *closed* under the operation of extracting roots. This means that for any z in **C**, $z^{1/n}$ is also in **C**. The real number system does not possess a similar closure property since, if x is in **R**, $x^{1/n}$ is not necessarily in **R**.

(*ii*) Geometrically, the n nth roots of a complex number z can also be interpreted as the vertices of a regular polygon with n sides that is inscribed within a circle of radius $\sqrt[n]{r}$ centered at the origin. You can see the plausibility of this fact by reinspecting Figures 1.4.1 and 1.4.2. See Problem 19 in Exercises 1.4.

(*iii*) When m and n are positive integers with no common factors, then (4) enables us to define a **rational power** of z, that is, $z^{m/n}$. It can be shown that the set of values $(z^{1/n})^m$ is the same as the set of values $(z^m)^{1/n}$. This set of n common values is defined to be $z^{m/n}$. See Problems 25 and 26 in Exercises 1.4.

EXERCISES 1.4

Answers to selected odd-numbered problems begin on page ANS-2.

In Problems 1–14, use (4) to compute all roots. Give the principal nth root in each case. Sketch the roots $w_0, w_1, \ldots, w_{n-1}$ on an appropriate circle centered at the origin.

1. $(8)^{1/3}$

2. $(-1)^{1/4}$

3. $(-9)^{1/2}$

4. $(-125)^{1/3}$

5. $(i)^{1/2}$

6. $(-i)^{1/3}$

7. $(-1+i)^{1/3}$

8. $(1+i)^{1/5}$

9. $(-1+\sqrt{3}i)^{1/2}$

10. $(-1-\sqrt{3}i)^{1/4}$

11. $(3+4i)^{1/2}$

12. $(5+12i)^{1/2}$

13. $\left(\dfrac{16i}{1+i}\right)^{1/8}$

14. $\left(\dfrac{1+i}{\sqrt{3}+i}\right)^{1/6}$

15. (**a**) Verify that $(4+3i)^2 = 7+24i$.

(**b**) Use part (a) to find the two values of $(7+24i)^{1/2}$.

16. Rework Problem 15 using (4).

17. Find all solutions of the equation $z^4 + 1 = 0$.

18. Use the fact that $8i = (2+2i)^2$ to find all solutions of the equation

$$z^2 - 8z + 16 = 8i.$$

The n distinct ***n*th roots of unity** are the solutions of the equation $w^n = 1$. Problems 19–24 deal with roots of unity.

19. (**a**) Show that the n nth roots of unity are given by

$$(1)^{1/n} = \cos\frac{2k\pi}{n} + i\sin\frac{2k\pi}{n}, \quad k = 0, 1, 2, \ldots, n-1.$$

(**b**) Find n nth roots of unity for $n = 3$, $n = 4$, and $n = 5$.

(**c**) Carefully plot the roots of unity found in part (b). Sketch the regular polygons formed with the roots as vertices. [*Hint*: See (*ii*) in the *Remarks*.]

20. Suppose w is a cube root of unity corresponding to $k = 1$. See Problem 19(a).

(**a**) How are w and w^2 related?

(**b**) Verify by direct computation that

$$1 + w + w^2 = 0.$$

(**c**) Explain how the result in part (b) follows from the basic definition that w is a cube root of 1, that is, $w^3 = 1$. [*Hint*: Factor.]

21. For a fixed n, if we take $k = 1$ in Problem 19(a), we obtain the root

$$w_n = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}.$$

Explain why the n nth roots of unity can then be written

$$1, w_n, w_n^2, w_n^3, \ldots, w_n^{n-1}.$$

22. Consider the equation $(z + 2)^n + z^n = 0$, where n is a positive integer. By any means, solve the equation for z when $n = 1$. When $n = 2$.

23. Consider the equation in Problem 22.

(**a**) In the complex plane, determine the location of all solutions z when $n = 5$. [*Hint*: Write the equation in the form $[(z + 2)/(-z)]^5 = 1$ and use part (a) of Problem 19.]

(**b**) Reexamine the solutions of the equation in Problem 22 for $n = 1$ and $n = 2$. Form a conjecture as to the location of all solutions of $(z + 2)^n + z^n = 0$.

24. For the n nth roots of unity given in Problem 21, show that

$$1 + w_n + w_n^2 + w_n^3 + \cdots + w_n^{n-1} = 0.$$

[*Hint*: Multiply the sum $1 + w_n + w_n^2 + w_n^3 + \cdots + w_n^{n-1}$ by $w_n - 1$.]

Before working Problems 25 and 26, read (*iii*) in the *Remarks*. If m and n are positive integers with no common factors, then the n values of the rational power $z^{m/n}$ are

$$w_k = \sqrt[n]{r^m}\left[\cos\frac{m}{n}(\theta + 2k\pi) + i\sin\frac{m}{n}(\theta + 2k\pi)\right], \tag{5}$$

$k = 0,\ 1,\ 2,\ \ldots,\ n - 1$. The w_k are the n distinct solutions of $w^n = z^m$.

25. (**a**) First compute the set of values $i^{1/2}$ using (4). Then compute $(i^{1/2})^3$ using (9) of Section 1.3.

(**b**) Now compute i^3. Then compute $(i^3)^{1/2}$ using (4). Compare these values with the results of part (b).

(**c**) Lastly, compute $i^{3/2}$ using formula (5).

26. Use (5) to find all solutions of the equation $w^2 = (-1 + i)^5$.

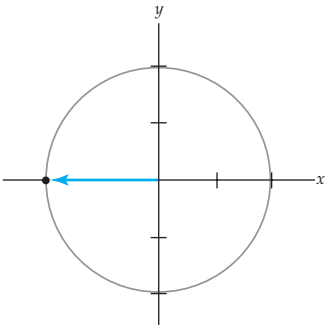


FIGURE 1.4.3 Figure for Problem 27

Focus on Concepts

27. The vector given in **FIGURE 1.4.3** represents one value of $z^{1/n}$. Using only the figure and trigonometry—that is, do not use formula (4)—find the remaining values of $z^{1/n}$ when $n = 3$. Write the values of $z^{1/n}$ in the form $a + ib$. Repeat for $n = 4$.

28. Suppose n denotes a nonnegative integer. Determine the values of n such that $z^n = 1$ possesses only real solutions. Defend your answer with sound mathematics.

29. (**a**) Proceed as in Example 2 to find the approximate values of the two square roots w_0 and w_1 of $1 + i$.

(**b**) Show that the exact values of the roots in part (a) are

$$w_0 = \sqrt{\frac{1+\sqrt{2}}{2}} + i\sqrt{\frac{\sqrt{2}-1}{2}}, \quad w_1 = -\sqrt{\frac{1+\sqrt{2}}{2}} - i\sqrt{\frac{\sqrt{2}-1}{2}}.$$

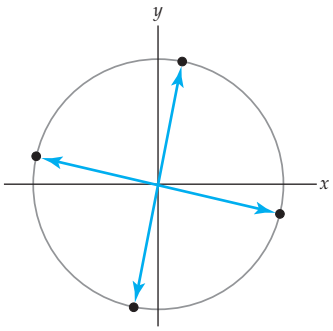


FIGURE 1.4.4 Figure for Problem 34

30. Discuss: What geometric significance does the result in Problem 24 have?

31. Discuss: A real number can have a complex nth root. Can a nonreal complex number have a real nth root?

32. Suppose w is located in the first quadrant and is a cube root of a complex number z. Can there exist a second cube root of z located in the first quadrant? Defend your answer with sound mathematics.

33. Suppose z is a complex number that possesses a fourth root w that is neither real nor pure imaginary. Explain why the remaining fourth roots are neither real nor pure imaginary.

34. Discuss: The vectors in FIGURE 1.4.4 represent the four fourth roots of a complex number z. Can you determine from this figure alone which root is the principal fourth root of z?

Computer Lab Assignments

In Problems 35–40, use a CAS to first find $z^n = w$ for the given complex number and the indicated value of n. Then, using the output and the same value of n, determine whether $w^{1/n} = (z^n)^{1/n} = z$. If not, explain why not.

35. $z = 2.5 - i;\ n = 10$

36. $z = -0.5 + 0.3i;\ n = 5$

37. $z = 1 + 3i;\ n = 8$

38. $z = 2 + 2i;\ n = 12$

39. $z = i;\ n = 21$

40. $z = -1 + \sqrt{3}\,i;\ n = 11$

1.5 Sets of Points in the Complex Plane

INTRODUCTION In the preceding sections we examined some rudiments of the algebra and geometry of complex numbers. But we have barely scratched the surface of the subject known as complex analysis; the main thrust of our study lies ahead. Our goal in the chapters that follow is to examine functions of a single complex variable $z = x + iy$ and the calculus of these functions.

Before introducing the notion of a function in Chapter 2, we need to state some essential definitions and terminology about sets in the complex plane.

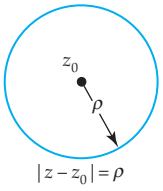


FIGURE 1.5.1 Circle of radius ρ

Circles Suppose $z_0 = x_0 + iy_0$. Since $|z - z_0| = \sqrt{(x - x_0)^2 + (y - y_0)^2}$ is the distance between the points $z = x + iy$ and $z_0 = x_0 + iy_0$, the points $z = x + iy$ that satisfy the equation

$$|z - z_0| = \rho, \quad \rho > 0, \tag{1}$$

lie on a **circle** of radius ρ centered at the point z_0. See FIGURE 1.5.1.